LEARNING DESIGN TOOLKIT (M.ED. IN IDET)

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Learning Design Toolkit

Learning Theories

Module 1

Behaviorism

Behavioral theory focuses on the way humans learn through interaction with their environment. Behaviorism solidifies the fact that behaviors are learned through conditioning. Skinner mentioned that conditioning is a procedure that utilizes punishment and reinforcement. The behavioral model provides an organized approach to teaching, allowing educators to set clear expectations and provide consistent routines. Behavioral theorists believe there is a certain way to do tasks to cultivate a desired outcome, and the teacher determines what that looks like (Samoila et al., 2023).

Example

This theory can be applied to student behavior plans. Many students thrive with positive reinforcement. They can better interact in school when their positive behaviors are rewarded (Bright, 2023). For example, if a young student struggles with speaking out inappropriately in class, they may have a positive behavior plan to encourage them when they appropriately follow classroom expectations. The plan could state that when the student does the desired task of raising their hand before asking questions, the teacher will thank them for raising their hand and waiting to be called on before speaking.

Reference(s)

Bright, K. (2023, September 21). *Behaviorism in education: How to foster learning environments.* LearnLever. https://learnlever.com/behaviorism-in-education/

Samoila, C., Ursutiu, D., & Munteanu, F. (2023). The remote experiment in the light of the learning theories. *International Journal of Online & Biomedical Engineering*, *19*(14), 26–44. https://doi.org/10.3991/ijoe.v19i14.43163

Cognitivism

Cognitivism is an objectivist learning theory, based on the learner acquiring an objective “truth” (Heaster-Ekholm, 2020, p. 53). Cognitivists focus on the active role of the learner. Their goal is to help individuals process and recall the knowledge outlined by learning objectives (Ertmer & Newby, 2013, pp. 50–54). Educators support this process by breaking information into manageable chunks, using strategies such as analogies and mnemonics, and providing practice and feedback to strengthen understanding (American College of Education [ACE], 2025, 2:40; Ertmer & Newby, 2013, p. 54).

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Example

In music, cognitivism can be applied to subjects such as music theory and notation. While musical expression allows for individual interpretation, established conventions, such as the names of notes on the staff, become objective knowledge to acquire. An educator might break notation into smaller sections and introduce mnemonics such as “FACE” for the treble clef spaces and “All Cows Eat Grass” for the bass clef spaces. Students then practice identifying notes, connecting them to their corresponding keys on the piano, and receiving corrective feedback. These strategies help students process, organize, retain, and recall musical information, providing a foundation for more advanced learning and creative expression.

Reference(s)

American College of Education. (2025). *TECH5203 Fundamentals of learning design and technology: Module 1* [Part 2 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378077

Ertmer, P. A., & Newby, T. J. (2013). Behaviorism, cognitivism, constructivism: Comparing critical features from an instructional design perspective. *Performance Improvement Quarterly, 26*(2), 43–71. https://doi.org/10.1002/piq.21143

Heaster-Ekholm, K. L. (2020). Popular instructional design models: Their theoretical roots and cultural considerations. *International Journal of Education and Development Using Information and Communication Technology, 16*(3), 50–65. https://eric.ed.gov/?id=EJ1275582

Constructivism

Constructivism is a learner-centered theory based on the belief that knowledge is actively constructed by the learner from their experiences, prior knowledge, beliefs, cultural backgrounds, and interactions with others (Ertmer & Newby, 2013, pp. 55–57; (Heaster-Ekholm, 2020, pp. 53–54). Instructors act as facilitators, guiding students through meaningful experiences, exploration, collaboration, and reflection rather than dictating how they should learn. Students take an active role in shaping their learning goals, evaluating their progress, and refining their understanding over time (American College of Education [ACE], 2025, 3:17; Ertmer & Newby, 2013, pp. 57–59; Heaster-Ekholm, 2020, pp. 53-54).

Example

Music and student expression go hand in hand, making constructivism applicable to composition and improvisation. The student draws on their prior musical knowledge, personal experiences, and creativity to develop their own song idea. The instructor gently guides the student through questions and encourages exploration. The student can then reflect on their composition and make changes based on their developing understanding. This allows the student to actively construct musical knowledge through meaningful experiences rather than simply following instructions.

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Reference(s)

American College of Education. (2025). *TECH5203 Fundamentals of learning design and technology: Module 1* [Part 2 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378077

Ertmer, P. A., & Newby, T. J. (2013). Behaviorism, cognitivism, constructivism: Comparing critical features from an instructional design perspective. *Performance Improvement Quarterly, 26*(2), 43–71. https://doi.org/10.1002/piq.21143

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Connectivism

Connectivism holds that learning is as much an external process as an internal one, spanning vast knowledge networks (Kizito, 2016). In connectivism, learning occurs as learners build connections with others through online collaboration (Jirasatjanukul & Jeerungsuwan, 2018). This builds a learning community in which learners contribute to one another's knowledge (American College of Education [ACE], 2025). As in constructivism, students' knowledge is shaped by personal experiences (Jirasatjanukul & Jeerungsuwan, 2018; Kizito, 2016). However, connectivism emphasizes learning through large knowledge networks, where students can encounter different perspectives and integrate what they discover into their understanding of the world (Jirasatjanukul & Jeerungsuwan, 2018; Kizito, 2016). The instructor guides students toward their learning goals while encouraging the free exchange of ideas, clarifying misunderstandings, and helping students navigate differing perspectives without tightly controlling the discussion.

Example

The instructor can encourage students to explore musical traditions from around the world through online performance videos, musician interviews, and educational resources. By examining multiple sources, students encounter different cultural perspectives and learn from musicians' firsthand experiences. Afterward, they share their discoveries through discussions or musical demonstrations. Through these exchanges, students consider different viewpoints and expand their understanding of music by learning from their peers' experiences with the topic. The instructor facilitates these discussions, clarifies misunderstandings, and encourages the exploration of diverse perspectives without prescribing a single interpretation.

Reference(s)

American College of Education. (2025). *TECH5203 Fundamentals of learning design and technology: Module 1* [Part 2 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378077

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Pedagogy vs. Andragogy

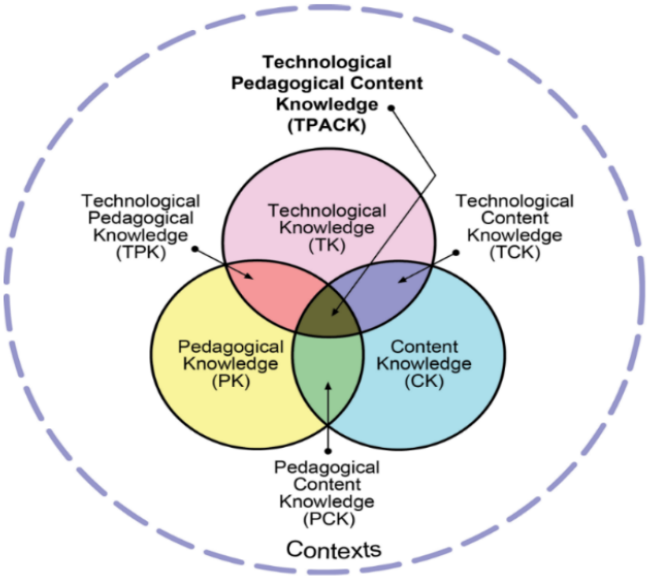
TPACK | Technological Pedagogical Content Knowledge

The TPACK model focuses on three types of knowledge an instructor should have. Technological Knowledge (TK) involves understanding how to select and use technology for instruction. Pedagogical Knowledge (PK) involves understanding how learning occurs and applying that knowledge to design effective instruction. Lastly, Content Knowledge refers to knowledge of the subject being taught (American College of Education [ACE], 2024; Tagayan, 2026). In addition, there are three more types of specialized knowledge that each combine aspects of the other two. First is Technological Pedagogical Knowledge, which concerns understanding how learning theories apply to teaching with various technologies. The second is Technological Content Knowledge, which involves knowing which technologies can represent, transform, support, or even constrain teaching of the subject matter at hand. Finally, Pedagogical Content Knowledge refers to knowing how pedagogical theory applies to a specific field of teaching (ACE, 2024; Alhamid & Mohammad-Salehi, 2024). At the center is TPACK, representing the integration of all three foundational knowledge domains and their intersections. Figure 1 illustrates how Technological Knowledge, Pedagogical Knowledge, and Content Knowledge intersect within the TPACK framework.

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Figure 1

Technological Pedagogical Content Knowledge (TPACK) Framework



Note. From Using the *TPACK Image*, by TPACK.org, n.d. (https://tpack.org/tpack-image/). Reproduced by permission of the publisher, © 2012 by tpack.org.

Example

These can be applied specifically to piano. For TK, this would look like understanding how to operate DAWs, digital instruments, MIDI controllers, and music notation software. Pedagogical Knowledge (PK) involves understanding how to apply instructional and inclusive teaching strategies to support students' learning. CK in music requires knowing theory, composition, and effective sight-reading and performance techniques. The intersections of these three foundational domains form three additional knowledge domains: Technological Pedagogical Knowledge (TPK), Technological Content Knowledge (TCK), and Pedagogical Content Knowledge (PCK). For example, TPK in music involves understanding the best ways to support students' learning experience with music technology, as well as the features these tools should have to be accessible. TCK would be understanding how pitch, rhythm, note duration, and other theory concepts are represented, manipulated, and played back by students. An example could be understanding how a DAW visually represents music, recognizing that it does not represent traditional staff notation in the same way. Finally, Pedagogical Content Knowledge (PCK) involves understanding how to apply effective instructional strategies to teach specific musical concepts and skills. This could look like teaching composition by modeling a short musical phrase, breaking the composition process into manageable steps, and providing feedback to help students develop their musical ideas. For example, an instructor could integrate these domains by teaching students to compose a short musical phrase using a DAW. The instructor applies Content Knowledge (CK) to establish the musical objectives, Pedagogical Knowledge (PK) to scaffold the composition process and provide feedback, and Technological Knowledge (TK) to select and operate a DAW that supports students' learning and accessibility needs. Together, these domains create a TPACK-informed music lesson.

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Reference(s)

Alhamid, I. S. G., & Mohammad-Salehi, B. (2024). Examining EFL teachers' technological pedagogical content knowledge (TPACK) and their attitudes towards online teaching. *LEARN Journal: Language Education and Acquisition Research Network, 17*(1), 715–733. https://doi.org/10.14456/learnjournal.2024.29

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TPACK.org. (n.d.). *Using the TPACK image*. https://tpack.org/tpack-image/

TAWOCK | Technology Andragogy Work Content Knowledge

Technology Andragogy Work Content Knowledge (TAWOCK) expands upon TPACK by incorporating workplace knowledge and replacing its pedagogical emphasis with an andragogical emphasis on adult learners in vocational education (Arifin, Nurtanto, Priyatna, et al., 2020, pp. 789–790).

TAWOCK comprises four foundational knowledge domains and four overlapping domains that emerge from their integration (Arifin, Nurtanto, Warju, et al., 2020, p. 700).

The following definitions synthesize the descriptions provided by the American College of Education (ACE, 2024, 5:34) and Arifin, Nurtanto, Kholifah, et al. (2020, p. 446). Table 1 summarizes the four foundational and four overlapping knowledge domains of TAWOCK.

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Table 1

Eight Knowledge Domains of TAWOCK

Domain category	Knowledge domain	Definition
Four foundational domains	Technology Knowledge (TK)	Understanding how to use technology to support learning.
	Andragogy Knowledge (AK)	Understanding adult learning approaches that promote independence.
	Content Knowledge (CK)	Understanding the subject matter being taught.
	Work Knowledge (WK)	Understanding workplace requirements and job-specific competencies.
Four overlapping domains	Technology Andragogy Knowledge (TAK)	Using technology to support adult learning.
	Technology Work Knowledge (TWK)	Understanding and integrating workplace technologies into instruction.
	Andragogy Content Knowledge (ACK)	Applying adult learning approaches to teach subject matter.
	Work Content Knowledge (WCK)	Aligning subject matter with workplace competencies.

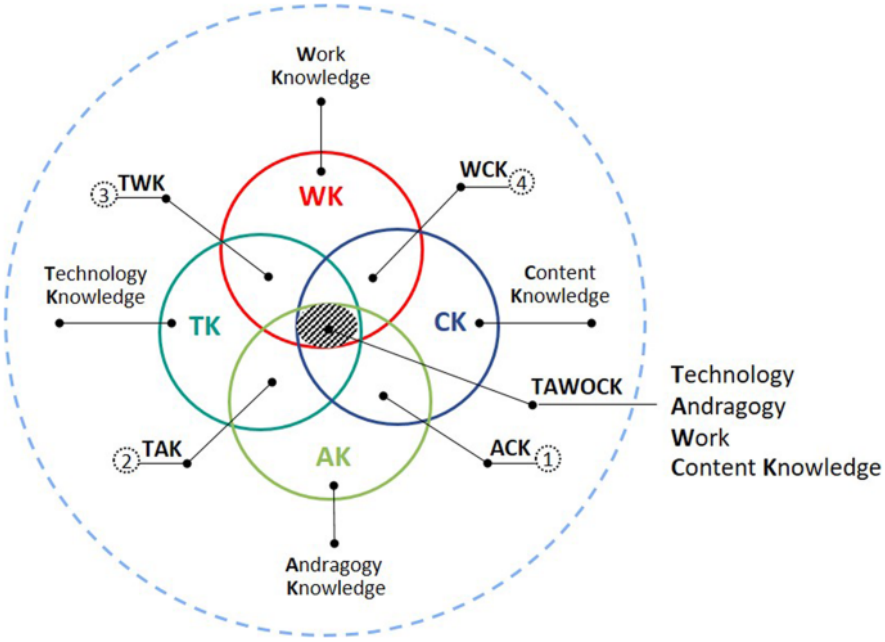
Note. Definitions synthesized from American College of Education (2024, p. 12) and Arifin, Nurtanto, Kholifah, et al. (2020, p. 446).

Together, these domains are intended to help educators design technology-supported learning experiences that encourage learner independence and develop the practical competencies needed for specific careers (Arifin, Nurtanto, Kholifah, et al., 2020, p. 446). Figure 2 illustrates how the foundational and overlapping knowledge domains are organized within the TAWOCK model.

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Figure 2

Technology Andragogy Work Content Knowledge (TAWOCK) Model



Note. From "The Effect of a Training Program Based on the (TAWOCK) Model for Teaching Computational Thinking Skills on Teaching Self-Efficacy Among Female Computer Teachers," by R. N. Alqahtani and A. Z. Almassaad, 2025, *Education and Information Technologies, 30,* p. 6693 (https://doi.org/10.1007/s10639-024-13088-x). Licensed under CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

Example

An example of TAWOCK is a professional development resource that helps music teachers evaluate and select technology for their studios. For this example, educators will evaluate a digital audio workstation (DAW) to determine whether it can help students compose and develop short musical pieces.

Technology Knowledge (TK): Educators explore the DAW's capabilities and limitations, including recording, MIDI input, editing, and playback.

Andragogy Knowledge (AK): Educators draw upon their prior teaching experiences, identify their individual professional needs, and independently determine what they hope to gain from the training.

Work Knowledge (WK): Educators consider the practical requirements of their studios, including cost, accessibility, device compatibility, and students' individual needs.

Content Knowledge (CK): Educators apply their understanding of musical concepts, including melody, harmony, rhythm, and composition.

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The four overlapping domains also apply to this professional development experience.

Technology Andragogy Knowledge (TAK): Educators independently explore the DAW through hands-on activities, drawing upon their existing technological experiences and learning at their own pace.

Technology Work Knowledge (TWK): Educators evaluate whether the DAW's features, accessibility options, and device requirements are compatible with the practical demands of their studios.

Andragogy Content Knowledge (ACK): Educators draw upon their musical backgrounds and teaching experiences to explore different approaches to composition, sharing their knowledge and learning from one another's experiences.

Work Content Knowledge (WCK): Educators connect their understanding of musical composition to their professional responsibilities, such as teaching students to develop melodies, structure musical phrases, and refine their compositions.

Reference(s)

Alqahtani, R. N., & Almassaad, A. Z. (2025). The effect of a training program based on the (TAWOCK) model for teaching computational thinking skills on teaching self-efficacy among female computer teachers. *Education and Information Technologies, 30*(5), 6687–6705. https://doi.org/10.1007/s10639-024-13088-x

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Arifin, Z., Nurtanto, M., Priyatna, A., Kholifah, N., & Fawaid, M. (2020). Technology andragogy work content knowledge model as a new framework in vocational education: Revised technology pedagogy content knowledge model. *TEM Journal, 9*(2), 786–791. https://doi.org/10.18421/TEM92-48

Arifin, Z., Nurtanto, M., Warju, W., Rabiman, R., & Kholifah, N. (2020). The TAWOCK conceptual model at content knowledge for professional teaching in vocational education. *International Journal of Evaluation and Research in Education, 9*(3), 697–703. https://doi.org/10.11591/ijere.v9i3.20561

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Learning Process Models

Kolb's Experiential Learning Theory

Kolb's Experiential Learning Theory describes learning as a continuous, four-stage cycle in which learners develop knowledge and skills through experience, reflection, conceptual understanding, and experimentation (American College of Education [ACE], 2025, 0:35; Phoon & Phoon, 2026, p. 38).

Through four interconnected stages, learners have a Concrete Experience (CE). Reflective Observation (RO) of those experiences constructs meaning in the student's mind. They then develop that meaning into an Abstract Conceptualization (AC). Finally, they seek to apply their evolving understanding to new experiences. This is referred to as Active Experimentation (AE) (Javahery & Bavandi, 2026, p. 31).

When the outcomes of Active Experimentation become new Concrete Experiences, learners can reflect on the results, refine their understanding, and continue the learning cycle (Aguilar & Tafesse, 2026, pp. 4, 11).

Example

In my piano studio, a student learning a new piece begins with a Concrete Experience (playing it). During Reflective Observation, the student thinks back to playing and what could have been more expressive. In this way, the student develops an Abstract Conceptualization of how dynamics, articulation, and phrasing communicate emotion. The student then applies these concepts to a new song, actively experimenting with different articulation, dynamics, and other musical means of expression. This performance becomes a new Concrete Experience, allowing the student to evaluate their progress and continue refining their musical expression.

Reference(s)

American College of Education. (2025). *TECH5203 Fundamentals of learning design and technology: Module 1* [Part 4 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378077

Aguilar, M. P., & Tafesse, W. (2026). Virtual reality in undergraduate business education: How learning occurs across Kolb's experiential learning cycle. *Contemporary Educational Technology, 18*(3), Article ep665. https://doi.org/10.30935/cedtech/18729

Javahery, P., & Bavandi, Z. (2026). Kolb's experiential learning theory in action: Fostering empathy and practical skills in language teacher education. *Reflective Practice, 27*(1), 30–44. https://doi.org/10.1080/14623943.2025.2521108

Phoon, E., & Phoon, E. (2026). A deeper understanding: Create more meaningful learning experiences with DoE, Lego and David Kolb's learning theory. *Quality Progress, 59*(5), 36–41.

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Gagné's Nine Events of Instruction

Gagné's Nine Events of Instruction is a cognitive instructional design model that uses nine external instructional events to stimulate internal learning processes (American College of Education [ACE], 2025, 1:26; McNeill & Fitch, 2023, pp. 521–522; Zhu & St. Amant, 2010, pp. 341–342). Table 2 summarizes Gagné's Nine Events of Instruction.

Table 2

Gagné's Nine Events of Instruction

#	Instructional Event	Description
1	Gain Attention	Get learners interested in the lesson.
2	Inform Learners of Objectives	Tell learners what they are expected to achieve.
3	Stimulate Recall of Prior Learning	Connect new information to learners' existing knowledge and past experiences.
4	Present the Content	Introduce new information using organized instructional materials and explanations.
5	Provide Learning Guidance	Use instructional strategies that help learners process and remember information.
6	Elicit Performance	Provide activities that allow learners to practice and demonstrate their new knowledge or skills.
7	Provide Feedback	Identify mistakes and misunderstandings and provide guidance to help learners improve.
8	Assess Performance	Evaluate learners' performance against the stated learning objectives.
9	Enhance Retention and Transfer	Reinforce learning and provide opportunities for learners to apply their knowledge and skills in new situations.

Note. Descriptions summarized from McNeill and Fitch (2023, pp. 522–523).

Together, these events provide a structured approach to instruction that helps learners build upon existing knowledge, develop new skills, and transfer their learning to different contexts, including real-world applications (Zhu & St. Amant, 2010, pp. 341–342, 358–359). However, the events do not always need to occur in a fixed order, nor must all nine be included for learning to take place (McNeill & Fitch, 2023, p. 522).

Example

In a virtual piano lesson, I could apply Gagné's Nine Events of Instruction to help a beginner identify individual piano notes by ear, starting with C, D, and E. Table 3 illustrates how Gagné's Nine Events of Instruction can be applied to teaching piano note identification by ear.

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Table 3

Application of Gagné's Nine Events of Instruction to Piano Note Identification by Ear

#	Instructional Event	Example
1	Gain Attention	Play a familiar melody and ask the student whether they can recognize any of its notes just by listening.
2	Inform Learners of Objectives	Explain that the student will learn to identify C, D, and E by ear without looking at the keyboard.
3	Stimulate Recall of Prior Learning	Review the notes C, D, and E on the keyboard, including their names and relationships to one another.
4	Present the Content	Play each note individually, identifying its name and allowing the student to become familiar with its sound.
5	Provide Learning Guidance	Provide C as a reference pitch and demonstrate how the student can compare each note with it to identify D and E.
6	Elicit Performance	Play individual notes without showing the keyboard and ask the student to identify each one.
7	Provide Feedback	Confirm correct answers and help the student compare incorrectly identified notes with the reference pitch.
8	Assess Performance	Play a new sequence of individual notes and evaluate how accurately the student identifies them independently, using the reference pitch.
9	Enhance Retention and Transfer	Revisit the activity in future lessons and gradually introduce additional notes, applying the skill to identifying notes in simple melodies.

Reference(s)

American College of Education. (2025). *TECH5203 Fundamentals of learning design and technology: Module 1* [Part 4 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378077

McNeill, L., & Fitch, D. (2023). Microlearning through the lens of Gagné's nine events of instruction: A qualitative study. *TechTrends, 67*(3), 521–533. https://doi.org/10.1007/s11528-022-00805-x

Zhu, P., & St. Amant, K. (2010). An application of Robert Gagné's nine events of instruction to the teaching of website localization. *Journal of Technical Writing and Communication, 40*(3), 337–362. https://doi.org/10.2190/TW.40.3.f

Continuous Improvement

PDSA Cycle | Plan Do Study Act Cycle

The Plan-Do-Study-Act (PDSA) Cycle helps educators continuously improve their courses through repeated cycles of evaluation and refinement (Gazza, 2015, pp. 291, 296). This model encourages testing smaller changes before fully implementing them (Zann et al., 2021, pp. e25–e26). Educators can collect quantitative and qualitative data and use the results to make

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decisions about which parts of their curriculum work and which still need improvement (American College of Education [ACE], 2024, 2:37).

The cycle consists of four repeating stages:

1. **Plan:** This is where the instructor identifies an area for improvement, establishes measurable goals, develops a plan, and determines how to evaluate success (ACE, 2024, 2:09; Gazza, 2015, p. 292; Zann et al., 2021, p. e25).
2. **Do:** This is where the teacher implements the planned change. Ideally, this will start on a small scale, while collecting data and recording observations (ACE, 2024, 2:21; Zann et al., 2021, p. e26).
3. **Study:** In this step, the teacher analyzes the results, compares them with the original goals and predictions, and identifies intended and unintended outcomes (ACE, 2024, 2:34; Zann et al., 2021, pp. e25–e26).
4. **Act:** Finally, they decide whether to adopt, modify, or replace the change based on the findings (ACE, 2024, 3:38). Then they use what was learned to repeat the next cycle of improvement (Zann et al., 2021, p. e30).

When using PDSA, instructors should treat instructional design as an ongoing cycle. Educators should continuously reexamine their curriculum to further refine their instructional approaches based on what they learn (Gazza, 2015, p. 296).

Example

An educator uses a technology evaluation rubric (Simko, 2026) to determine whether a new digital learning tool meets students' instructional and accessibility needs.

1. **Plan:** Identify an instructional need, select a potential technology tool, and establish evaluation criteria using the rubric. Set a measurable goal, such as having at least 80% of students independently complete a designated learning activity.
2. **Do:** Pilot the tool with a small group of students, recording whether they independently complete the designated learning activity while collecting feedback and observing their experiences.
3. **Study:** Analyze rubric results and compare student performance against the established goal. Identify strengths, limitations, and unexpected challenges.
4. **Act:** Use the rubric findings to determine whether the tool is a Strong Choice, Worth Considering, or Not a Strong Choice (Simko, 2026). Decide whether to adopt, adapt, investigate further, or select an alternative. Repeat the cycle to evaluate its effectiveness over time.

Reference(s)

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 1* [Part 5 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378077

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Zann, A., Harwayne-Gidansky, I., & Maa, T. (2021). Incorporating simulation into your Plan-Do-Study-Act cycle. *Pediatric Annals, 50*(1), e25–e31. https://doi.org/10.3928/19382359-20201213-01

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Design Models

Module 2

Instructional Design (ID) vs. Learning Experience Design (LXD)

Instructional Design (ID) and Learning Experience Design (LXD) define an approach to creating a learning experience. Design includes understanding the audience's needs and goals to create learning activities and effective content delivery. The two methods are complementary and do not have to exist exclusively. ID emphasizes learning and supports a step-by-step process for developing instruction. Typically, instruction is driven by learning objectives, direct instruction, and uses traditional assessments to evaluate learning. LXD has an integrated approach with a heavier emphasis on emotional, social, and cognitive aspects of learning. In LXD, the focus is on integrating hands-on and collaborative instruction that often utilizes technology. According to Floor (2023), “A great way to explain the general difference between LXD and ID is by comparing a scientist to an artist” (para. 2). Like a scientist, the instructional designer follows a methodical process with clear objectives and measurable results. Meanwhile, the learning experience designer, like an artist, creates engaging, emotionally rich experiences that connect with learners on multiple levels beyond just transferring information.

Example

An example demonstrating the difference between ID and LXD can be illustrated in a professional development session with school staff. The staff will receive training on the new Assessment Feature in the school’s LMS. Table 4 compares how Instructional Design (ID) and Learning Experience Design (LXD) can be applied to training school staff to use a new LMS assessment feature.

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Table 4

Comparison of Instructional Design and Learning Experience Design in LMS Training

Instructional Design (ID)	Learning Experience Design (LXD)
The staff is presented with the learning objective to learn the new online feature. Staff are required to use online assessments to collect data. ↓ The instructor demonstrates the skills and steps to complete the goals. ↓ The staff members try the step-by-step implementation as guided by the instructor. ↓ Staff have begun using the new online feature in their classrooms. Instructors and administration evaluate and hold staff accountable for carrying out the directive.	The staff is presented with a problem that needs to be solved. The administration needs a way to collect data as students complete formative work. The staff are given the new online assessment feature on the LMS as a possible option. ↓ The staff begins to explore the LMS and its features to experiment with how it works and what it can do. As they explore, they create lists of pros and cons. The staff notes items for which they need further training or exploration. ↓ Through collaboration, the staff determines that the online LMS is the best way to collect data. The staff begins using the assessment feature to collect data. ↓ As the data are collected, the administration and staff evaluate the effectiveness and adjust accordingly.

Note. ID = instructional design; LXD = learning experience design; LMS = learning management system. The examples compare two approaches to training staff to use an online assessment feature.

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First Principles of Instruction

Merrill's Principles of Instruction

Merrill's First Principles of Instruction identify five foundational conditions that promote effective learning, regardless of the instructional model or method used (Merrill, 2002, pp. 43–45). Merrill proposed that incorporating these principles into instruction would improve learning and performance (Merrill, 2002, p. 44). Table 5 summarizes Merrill's First Principles of Instruction.

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Table 5

Merrill's First Principles of Instruction

#	Principle	Description
1	Problem-Centered	Learning is promoted when learners engage with meaningful, real-world problems or tasks.
2	Activation	Learning is promoted when learners connect new information to relevant prior knowledge, experiences, or existing mental structures.
3	Demonstration	Learning is promoted when new knowledge or skills are demonstrated to learners in ways that align with the learning goal, rather than simply being described.
4	Application	Learning is promoted when learners practice and apply new knowledge or skills while receiving appropriate guidance and feedback.
5	Integration	Learning is promoted when learners connect new knowledge to their own lives through reflection, demonstration, discussion, or creating new ways to use what they have learned.

Note. Descriptions summarized from ACE (2024) and Merrill (2002).

Example

Instructors can apply Merrill's First Principles of Instruction in piano education when teaching a student how to use the sustain pedal effectively. The goal is for the student to recognize when pedaling improves the sound, change the pedal at appropriate times, and avoid creating blurred harmonies. Table 6 illustrates how Merrill's First Principles of Instruction can be applied to teaching sustain pedal technique.

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Table 6

Application of Merrill's First Principles of Instruction to Effective Sustain Pedal Use

#	Instructional Event	Example
1	Problem-Centered	The instructor plays a short passage with the sustain pedal held continuously, causing the harmonies to blur together. The student is challenged to determine how the pedal could be used to create a connected sound without allowing one harmony to interfere with the next.
2	Activation	The instructor asks the student to recall what they already know about how notes continue sounding after a key is released and how different harmonies can sound when played together. The student listens again and identifies places where the sound becomes unclear.
3	Demonstration	The instructor demonstrates how to depress and release the pedal as the harmony changes. They compare several versions of the same passage, including no pedal, continuously held pedal, and correctly changed pedal, so the student can hear the effect of each approach.
4	Application	While the student practices pedaling with the harmonic changes, the instructor provides feedback when the pedal changes too early, too late, or not at all, gradually allowing the student to make these adjustments independently.
5	Integration	The student demonstrates the pedaling technique for another person and explains why the pedal changes occur where they do. They then apply the same decision-making process to another passage that requires clean pedaling.

Pedaling gives the student a practical musical problem to solve rather than an isolated technique to memorize. The student draws on what they already know about sound and harmony, hears the technique demonstrated, practices it with feedback, and eventually applies the same decision-making process in another passage. The skill develops through the full progression from prior knowledge to independent musical use.

Reference(s)

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 2* [Part 3 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378082

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Merrill's Four-Phase Cycle of Instruction

Merrill's Four-Phase Cycle of Instruction illustrates how the First Principles of Instruction can be implemented through four interconnected phases within a problem-centered learning

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environment (American College of Education [ACE], 2024, 3:01; Merrill, 2002, p. 44). Within this problem-centered environment, learners move through the four phases shown in Table 7 as they work with increasingly complex tasks.

Table 7

Merrill's Four-Phase Cycle of Instruction

#	Principle	Description
1	Activation	Connect new learning to prior knowledge and experiences.
2	Demonstration	Show learners how to perform a skill or apply a concept through clear examples.
3	Application	Provide opportunities for learners to practice, receive guidance, and work through difficulties.
4	Integration	Encourage learners to incorporate their new knowledge and skills into real-world activities.

Note. Descriptions summarized from ACE (2024) and Merrill (2002).

Together, these phases provide a structured approach to implementing Merrill's First Principles rather than representing a separate instructional theory (ACE, 2024, 3:01; Merrill, 2002, p. 44).

Example

An example of Merrill's Four-Phase Cycle of Instruction in piano is teaching a student to transpose a familiar melody into another key. The goal is for the student to use their understanding of scale relationships and intervals to recreate the melody at a different starting pitch. Table 8 illustrates how Merrill's Four-Phase Cycle of Instruction can be applied to melodic transposition.

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Table 8

Application of Merrill's Four-Phase Cycle of Instruction to Transposition

#	Principle	Description
1	Activation	The instructor asks the student to recall major scales, whole and half steps, and the interval relationships between notes in a familiar melody. The student identifies the starting note and examines how the melody moves from one note to the next.
2	Demonstration	The instructor demonstrates how to move a short melodic passage into a different key while preserving the same note relationships. They think through the process aloud so the student can see how each new pitch is determined.
3	Application	The student transposes another section of the melody. At first, the instructor provides guidance when the student loses track of the interval pattern or key. As the student becomes more comfortable, they complete additional phrases with less assistance.
4	Integration	The student transposes the complete melody into a new key and performs both versions. They explain how they determined the new notes, then use the same process to independently transpose another familiar melody.

Transposition gives the student a chance to use scales and intervals in a practical way. The instructor first models how to move a familiar melody into a different key, and then the student works through the process with help as needed. By the end, the student is using the same approach on another melody without relying as heavily on the instructor.

Reference(s)

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 2* [Part 3 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378082

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Project Management Models

Agile

Agile design models use a flexible, iterative approach in which a project is developed and refined through repeated cycles rather than following a fixed plan from beginning to end. This allows designers to make incremental changes as the context, learner needs, or other project requirements change (American College of Education [ACE], 2024, 1:16; Grant & Kidwell, 2024, p. 213). Collaboration and user feedback are important to Agile design. Stakeholders remain involved in decisions as the project develops, and their feedback can lead to changes when new needs or information arise (Grant & Kidwell, 2024, pp. 213–214; Park, 2024, pp. 1, 3). In Agile design, the project is tested and revised as it develops. What is learned from each round can be

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used to make changes before moving forward (Lee & Chen, 2026, p. 759; Park, 2024, p. 4). Agile is particularly useful when a project may need to change after it begins. If learners' needs, feedback, or other conditions change, the designer can revise the next stage rather than sticking to the original plan (Lee & Chen, 2026, pp. 756–757).

One specific approach is Conrad Gottfredson's Agile instructional design model, which draws on Agile software development and project management. The model consists of five phases: Align, Get Set, Iterate and Implement, Leverage, and Evaluate (Drysdale, 2019, p. 59). Table 9 summarizes the five phases of Gottfredson's Agile instructional design model.

Table 9

Agile

#	Principle	Description
1	Align	Stakeholders and team members align the project with organizational strategy and need and determine the value it should provide to learners.
2	Get Set	The designers analyze the audience, performance goals, and operational requirements of the project before developing the instructional materials.
3	Iterate and implement	Instructional materials are developed and implemented through rapid cycles, allowing the learning experience to be adapted as development progresses.
4	Leverage	The organization uses its available technology resources to make the learning experience broadly accessible to learners.
5	Evaluate	The quality and success of the developed learning experience are evaluated.

Note. Descriptions summarized from Drysdale (2019).

The Agile model moves from alignment and planning into rapid development and implementation, then concludes with attention to access and evaluation (Drysdale, 2019, p. 59).

Example

Agile could be used to develop a practice-planning system that helps piano students organize their work between lessons. Rather than creating a single, complete system and expecting it to work for every student, the instructor can establish what the system needs to accomplish, develop it through repeated cycles, and evaluate how well it supports students' independent practice. Table 10 illustrates how Agile can be applied to developing a practice-planning system for piano students.

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Table 10

Application of Agile to a Practice-Planning System

#	Principle	Description
1	Align	The instructor identifies the need the practice-planning system should address. Some students may have difficulty deciding what to practice, how long to spend on different tasks, or what to do when they encounter a problem between lessons. The goal is to develop a system that helps students organize their practice and make more independent decisions.
2	Get Set	The instructor considers the students who will use the system, what they should be able to do with it, and the resource's practical requirements. For example, students may differ in age, reading ability, attention needs, practice routines, or access to technology. The instructor determines that the first version should include sections for practice goals, assigned activities, and questions students want to bring to their next lesson.
3	Iterate and implement	The instructor creates an initial version of the practice plan and tries it with a small group of students. Students are asked which sections they use, where they become confused, and whether the plan helps them make decisions during independent practice. The instructor revises the system based on their experiences and continues testing updated versions as the resource develops.
4	Leverage	The instructor uses technology already available to students to make the practice plan accessible between lessons. For example, students could access a digital version on the phone, tablet, or computer they normally use at home rather than needing a separate platform or device.
5	Evaluate	The instructor examines whether students are better able to decide what to practice, organize their practice time, and identify questions to bring to their next lesson. Student feedback and lesson observations can be used to determine the quality and success of the practice-planning system.

The practice-planning system begins with a specific need and an understanding of the students who will use it. Rather than waiting until the entire resource is finished, the instructor develops and implements it through repeated cycles, making changes as students use it. Providing access through technology students already have and evaluating whether the system improves independent practice carries the project through each phase of Agile.

Reference(s)

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 2* [Part 4 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378082

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SAM | Successive Approximation Model

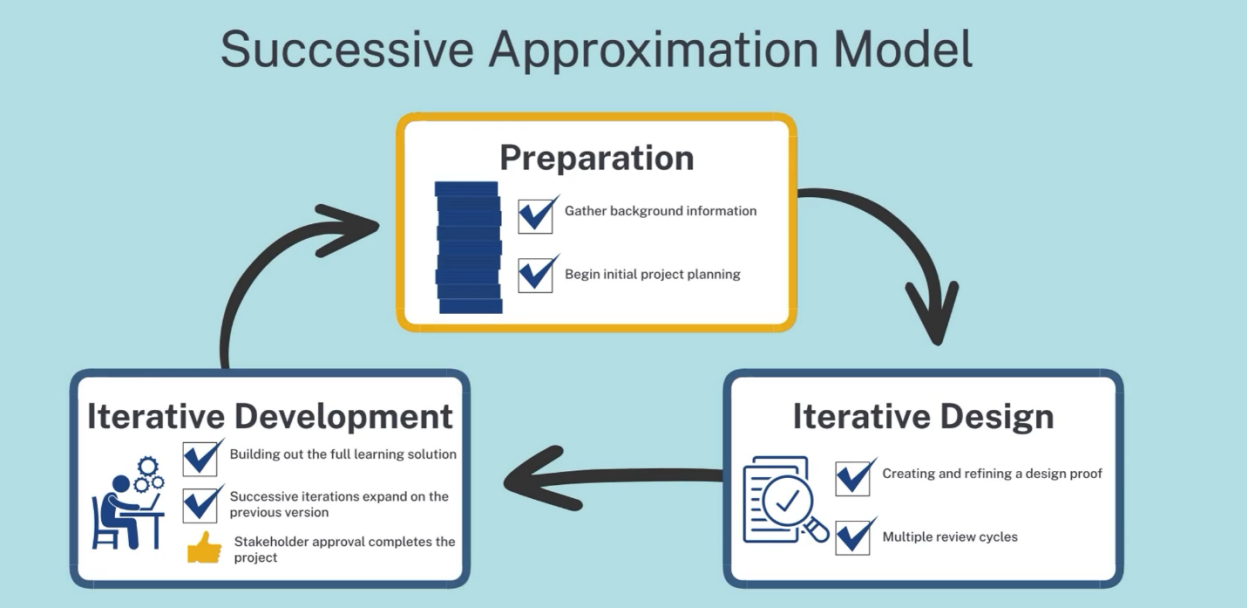
The Successive Approximation Model (SAM) is an agile, iterative approach to instructional design in which designers create early drafts or prototypes and improve them through repeated cycles of feedback, evaluation, and revision (American College of Education [ACE], 2024, 1:32; Tamez, 2016, pp. 19–20; Wolverton & Hollier, 2022, pp. 264–266). Rather than completing each stage once in a fixed sequence, SAM uses short design cycles that allow ideas to be tested and refined throughout development (Tamez, 2016, pp. 19–20). This specific model consists of three main phases: preparation, iterative design, and iterative development, moving from gathering information and planning to refining a design and ultimately developing a working solution to be tested again (ACE, 2024, 1:54). Because evaluation and revision occur throughout the process, SAM supports continuous improvement and allows a learning solution to continue evolving as

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needs or feedback change (Wolverton & Hollier, 2022, pp. 265–266, 270). Figure 3 illustrates the iterative relationship between preparation, iterative design, and iterative development.

Figure 3

Successive Approximation Model (SAM)



Note. Created by the author based on the Successive Approximation Model presented by ACE (2024, 6:55).

Example

An example of SAM is the development of an instructional resource that helps piano students understand and perform rhythms in compound meter. Rather than creating the entire resource before students use it, the instructor develops and tests small prototypes before expanding it. Table 11 illustrates how SAM can be applied to the development of an instructional resource for compound meter.

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Table 11

Application of SAM to an Instructional Resource That Helps Piano Students Understand and Perform Rhythms in Compound Meter

#	Principle	Description
1	Preparation	The instructor identifies where students commonly struggle with compound meter, such as feeling the larger beat, understanding how eighth notes are grouped, or distinguishing 6/8 from simple meters. The instructor also determines what students should be able to recognize and perform after using the resource.
2	Iterative Design	The instructor creates a small prototype that teaches one concept, such as feeling two large beats in 6/8. The prototype may include a visual beat diagram, a short clapping activity, and several rhythm examples. Next, the instructor tests the prototype with a few students and observes which explanations make sense, where students lose the beat, and which activities help them feel the grouping. The prototype is then revised and tested again.
3	Iterative Development	After refining the first activity, the instructor expands the resource to include additional compound-meter concepts, such as dotted-quarter-note beats, rhythmic subdivisions, and comparisons between 6/8 and simple meter. Students continue using the materials as they are developed, and their responses guide revisions to the directions, examples, and activities.
Cycle Back to Phase 1: Preparation		If testing reveals a new problem, such as students understanding the notation but still struggling to feel the larger pulse, the instructor returns to the preparation phase. Additional information is gathered about the difficulty, and another SAM cycle begins with a focus on that specific need.

The instructor first conducts a brief activity to help students feel the two main beats in 6/8. Student responses show whether the beat diagram, clapping activity, and rhythm examples are helping them understand the meter. Those results guide revisions before the instructor adds more difficult ideas, such as subdivisions and comparisons with simple meter. If students still struggle with a particular concept, that difficulty becomes the focus of the next round of planning and testing.

Reference(s)

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 2* [Part 4 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378082

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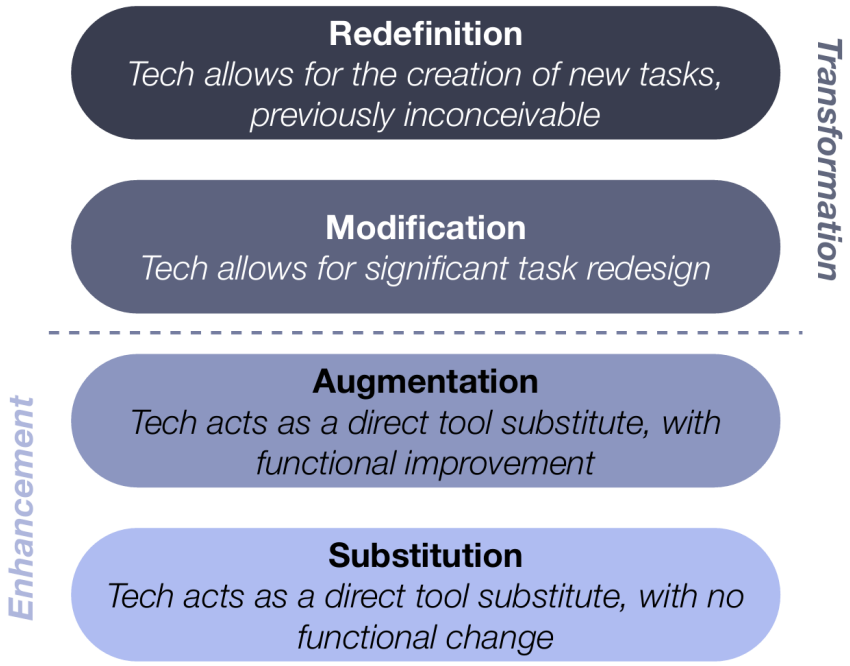
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SAMR | Substitution, Augmentation, Modification, and Redefinition

The SAMR model describes different ways technology can influence instruction, ranging from replacing an existing tool to enabling substantially different learning experiences (American College of Education [ACE], 2024, 4:33; Hamilton et al., 2016, p. 434). The model progresses from using technology as a replacement for an existing tool to using technology to create learning experiences that would not otherwise be possible. At the **substitution** level, technology directly replaces an existing tool without changing its function. **Augmentation** still involves replacement, but technology provides functional improvements. **Modification** uses technology to significantly redesign the same task, while **redefinition** uses technology to create new learning tasks (Hamilton et al., 2016, pp. 434–435). Figure 4 illustrates the four levels of the SAMR model, progressing from substitution to redefinition.

Figure 4

The SAMR Model



Note. From *Building transformation: An introduction to the SAMR model* (slide 2), by R. R. Puentedura, 2014, Hippasus. https://hippasus.com/blog/2014/08/22/building_transf_1/

Example

A piano student keeps a practice journal to document what they worked on, identify difficulties, and reflect on their progress. The instructor considers how technology could change this activity at progressively different levels of integration. Table 12 illustrates how the four levels of SAMR can be applied to piano practice journals.

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Table 12

Application of SAMR to Practice Journals and Reflective Writing

#	Principle	Description
1	Substitution	The student types their practice journal in a digital document instead of writing it in a paper notebook. The format changes, but the student records the same information in essentially the same way.
2	Augmentation	The student keeps the practice journal digitally and uses built-in features to make it easier to manage. For example, the student can check off completed assignments, set a reminder for the next practice session, and open a linked recording or score directly from the journal. The task is still to keep a practice journal, but the digital version makes the process more convenient and useful.
3	Modification	The student adds short audio or video recordings to journal entries and compares them with earlier recordings. The instructor can respond directly to specific entries or recordings, allowing the student to connect written reflection with evidence of their playing and use that feedback during later practice.
4	Redefinition	The student develops a multimedia record of their progress over an extended period by combining dated reflections, recordings, goals, feedback, and examples of revisions. They can move between earlier and later performances to identify patterns in their development and create a retrospective presentation explaining how their practice strategies changed and which approaches produced the strongest results.

The practice journal keeps the same general purpose while technology gradually changes what the student can do with it. At the lower levels, digital tools replace or improve the existing journal. At the higher levels, recordings, feedback, and multimedia reflection reshape the activity and eventually allow the student to analyze their development in ways that would be difficult to reproduce with a traditional written journal alone.

Reference(s)

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 2* [Part 4 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378082

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Waterfall

The Waterfall Design Model is a structured, sequential approach in which a project moves through clearly defined phases. (American College of Education [ACE], 2024; Saravanos, 2025, p. 1). In its traditional form, the project progresses from beginning to end in a mostly linear sequence, with one phase completed before moving to the next (ACE, 2024; Saravanos, 2025, pp. 1–2). Each stage leads into the next, creating an ordered process that moves from an initial goal or requirement through later design decisions (Ahern, 2016, p. 340). Although Waterfall is often described as strictly linear (Saravanos, 2025, pp. 1–2), some versions allow limited repetition or movement back to an earlier phase when problems are identified.

Example

The Waterfall model could be applied to planning a studio recital when major decisions must be made in a set order. Each phase depends on decisions made in the previous phase, so the project progresses from defining the event to carrying it out, then maintaining the materials afterward. Table 13 illustrates how the Waterfall model can be applied to planning a studio recital.

Table 13

Application of Waterfall to Recital Planning

#	Principle	Description
1	Requirements	The instructor determines the basic requirements of the recital, including the date, location, number of performers, available performance time, equipment needs, and any venue restrictions.
2	Design	Once the requirements are established, the instructor plans the recital's structure. This includes the performance order, approximate timing, seating arrangement, transitions between performers, and other logistical details.
3	Implementation	The instructor puts the plan into action by finalizing the program, preparing the venue, communicating expectations to students and families, and carrying out the recital according to the established schedule.
4	Verification	Once the recital is over, the instructor compares what was planned with what happened. If performers ran longer than expected or transitions took more time than planned, those differences are noted before the same recital format is used again.
5	Maintenance	The instructor updates the recital materials based on what happened during the event. For example, if the program ran behind because transitions took longer than expected, the timing guide can be revised before the next recital.

Planning a recital works well with a Waterfall structure because each stage relies on information established earlier. The event cannot be organized until its requirements are known, and the final plan cannot be evaluated until it has been carried out. Reviewing the results afterward also gives the instructor a way to improve the materials and procedures used for future recitals.

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Reference(s)

Ahern, T. C. (2016). A waterfall design strategy for using social media for instruction. *Journal of Educational Technology Systems, 44*(3), 332–345. https://doi.org/10.1177/0047239515615853

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 2* [Part 4 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378082

Saravanos, A. (2025). A brief history of the Waterfall Model: Past, present, and future. https://doi.org/10.48550/arXiv.2510.03894

ADDIE | Analysis, Design, Development, Implementation, and Evaluation

ADDIE is a widely used instructional design process that guides the development of instruction through five phases (Heaster-Ekholm, 2020, p. 54). Table 14 summarizes the five phases of the ADDIE instructional design process.

Table 14

ADDIE

#	Principle	Description
1	Analysis	The educator identifies what needs to be taught, the learners and their needs, and the goals or objectives of the instruction.
2	Design	In this phase, the objectives and tasks are matched with the content and assessments that will support them.
3	Development	The plans from the Design phase are then turned into the actual instructional materials, media, and assessments used in the course.
4	Implementation	The completed instruction is delivered to the intended learners.
5	Evaluation	The designer determines whether the instruction has met its intended goals and uses formative and summative evaluations to assess learning and the course's effectiveness.

Note. Descriptions summarized from ACE (2024) and Heaster-Ekholm (2020).

ADDIE was originally presented as a sequence of steps, but designers may return to an earlier phase when a problem comes up, or a change is needed (ACE, 2024, 2:34; Drljača et al., 2024, pp. 695–696; Heaster-Ekholm, 2020, p. 54). Evaluation is not limited to the end of the process. It can also be used during instruction design and implementation to identify what needs revision (ACE, 2024, 1:46). This gives designers a way to test and improve materials without abandoning ADDIE's overall sequence (Drljača et al., 2024, pp. 695–696).

Example

ADDIE could be used to develop a structured program that helps students build fluency with scales and arpeggios. The program would address the technical knowledge, movement patterns, and practice strategies students need to play them accurately and consistently. Table 15 illustrates how ADDIE can be applied to developing a structured program for scales and arpeggios.

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Table 15

Application of ADDIE to Core Music Skills

#	Principle	Description
1	Analysis	Determine what students already know about scales and arpeggios and identify common difficulties, such as fingering changes, uneven rhythm, uncertainty about key patterns, or difficulty coordinating the hands. The instructor also considers students' current technical level and any needs that could affect how the material is presented or practiced.
2	Design	Establish what students should be able to do by the end of the program and organize the skills into a logical sequence. For example, students might first learn the fingering and physical pattern for selected scales before increasing tempo, combining hands, and adding related arpeggios. The instructor also determines how progress will be assessed.
3	Development	Create instructional materials for the program, including fingering references, practice sequences, tempo-tracking sheets, demonstration videos, and short technical checkpoints. Materials can be adapted to improve readability or to provide alternative ways to access the instructions.
4	Implementation	The instructor introduces the scale and arpeggio program during regular lessons and has students begin using the assigned practice materials at home. For example, a student might spend the first week practicing one-octave scales using the planned fingering and tracking sheet, while the instructor listens during the next lesson to see where the sequence needs clarification or slowing down.
5	Evaluation	Review whether students are meeting the intended goals, which activities are helping them, and where they continue to struggle. Formative evaluation occurs throughout the program through lesson observations, technical checkpoints, and student feedback. For summative evaluation, the instructor can assess whether students can perform the selected scales and arpeggios accurately, with consistent fingering, rhythm, and technique at the established target tempo. The results can then be used to revise the sequence or materials before the program is used again.

The scale and arpeggio program moves from identifying learner needs to planning the instructional sequence, creating the necessary materials, using them with students, and evaluating the results. Each phase informs the next, while evaluation provides information to revise earlier decisions. This creates a structured development process while still allowing the program to improve after students begin using it.

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Reference(s)

American College of Education. (2024). *TECH5203 fundamentals of learning design and technology: Module 2* [Part 5 presentation]. Canvas. https://ace.instructure.com/courses/2217956/modules/items/48378082

Drljača, D. P., Tomić, S., & Ris, K. (2024). Benchmarking instructional design models - ADDIE wins. *Economy & Market Communication Review / Casopis Za Ekonomiju i Trzisne Komunikacije*, *14*(2), 688–698. https://doi.org/10.7251/EMC2402688D

Heaster-Ekholm, K. L. (2020). Popular instructional design models: Their theoretical roots and cultural considerations. *International Journal of Education and Development Using Information and Communication Technology, 16*(3), 50–65.

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Cognitive Load

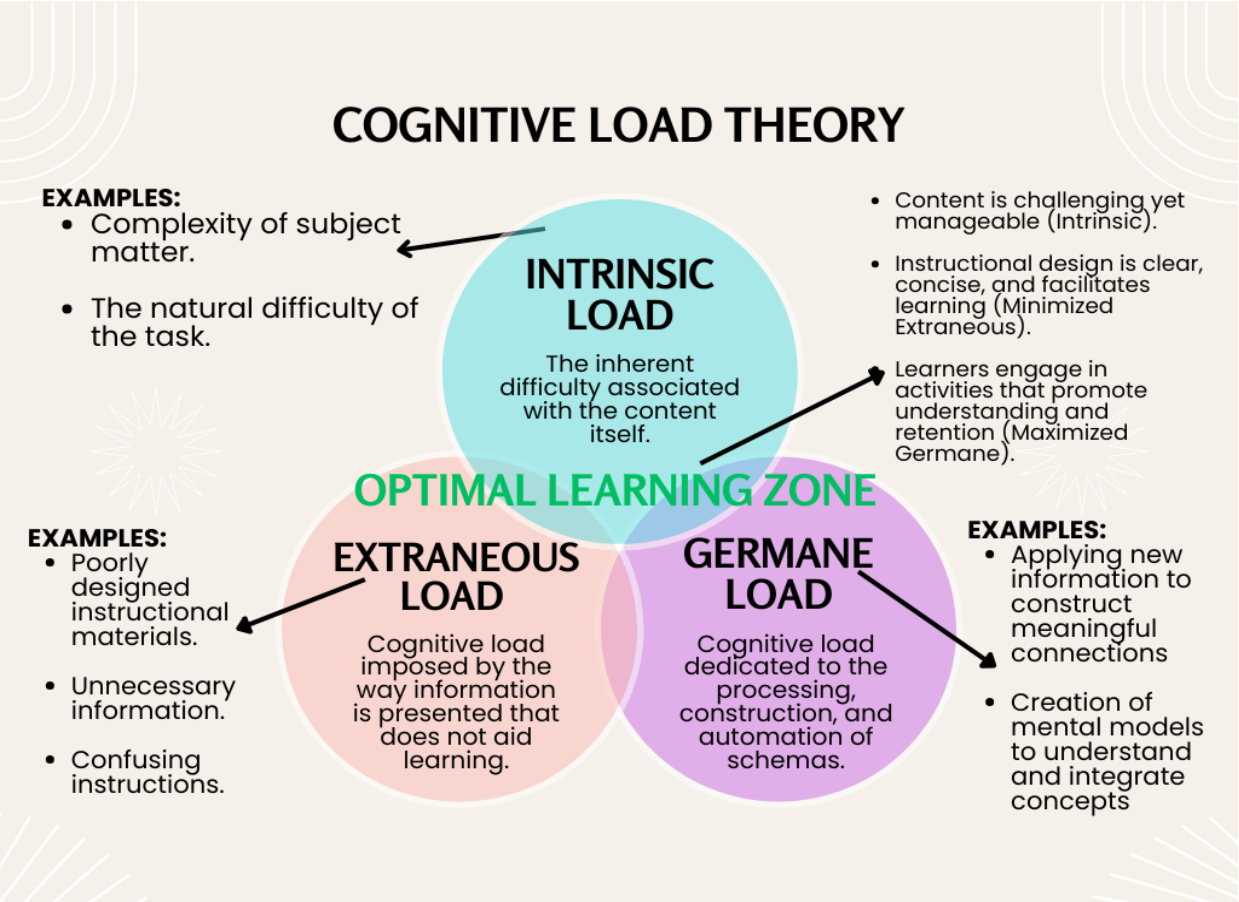
Module 3

Cognitive Load Theory

Cognitive load theory provides a framework for understanding the mental processes involved in learning and the limitations of our working memory. Clark and Mayer (2024) define learning as a change in knowledge caused by experience. Cognitive load theory builds on this by examining how our brains process, store, and manage information during learning. Figure 5 illustrates the relationships among intrinsic, extraneous, and germane cognitive load and an optimal learning zone.

Figure 5

Types of Cognitive Load



Note. The figure illustrates intrinsic, extraneous, and germane cognitive load, including examples of each and their relationship to an optimal learning zone. Graphic provided in the American College of Education Learning Design Toolkit course template.

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At its core, the theory recognizes that our working memory has limited capacity. When these limits are exceeded, learning becomes ineffective. The theory identifies three distinct types of cognitive load:

Intrinsic load represents the inherent complexity of the material being learned. This load varies based on the learner's prior knowledge and the complexity of the content itself. Effective instruction carefully manages this essential processing to enhance learning (Clark & Mayer, 2024).

Extraneous load is the unnecessary mental effort caused by poor instructional design. This includes confusing layouts, irrelevant information, or unnecessarily complex explanations. As Clark and Mayer (2024) emphasize, well-designed courses minimize extraneous processing through thoughtful design strategies.

Germane load is the productive mental effort that contributes to deeper understanding. This includes activities that help learners construct schemas and apply knowledge. Courses should intentionally foster this generative processing to maximize learning and create lasting memories (Clark & Mayer, 2024).

In simple terms, cognitive load can be understood as the total mental effort required by a learner's brain when processing new information. Instructional designers must carefully balance these three types of cognitive load—reducing extraneous load, managing intrinsic load, and optimizing germane load—to create effective learning experiences.

Example

Cognitive load theory would apply during any instructional design project. If instructors were designing online professional development for training on new gradebook software at a school, they would minimize unnecessary cognitive burdens because learners have limited working memory capacity (Sweller, 2020). The designer would manage the intrinsic load by referring to what the staff currently uses for a gradebook. The course would refer to terms and names that the staff knows and break each component into smaller bits of information. The designer would minimize extraneous load by using a variety of learning experiences. The necessary content would be concise, include well-organized images, highlight important steps, and show each step necessary. Staff could easily follow the steps needed to learn the new gradebook. The training would maximize germane load by asking staff to work in groups and practice what they are learning during the training. The professional development would include scenarios asking the staff to solve problems using the new gradebook.

Reference(s)

Clark, R. C., & Mayer, R. E. (2024). *E-learning and the science of instruction: Proven guidelines for consumers and designers of multimedia learning*. John Wiley & Sons, Inc.

Sweller, J. (2020). Cognitive load theory and educational technology. *Educational Technology Research and Development, 68*(1), 1–16. https://doi.org/10.1007/s11423-019-09701-3

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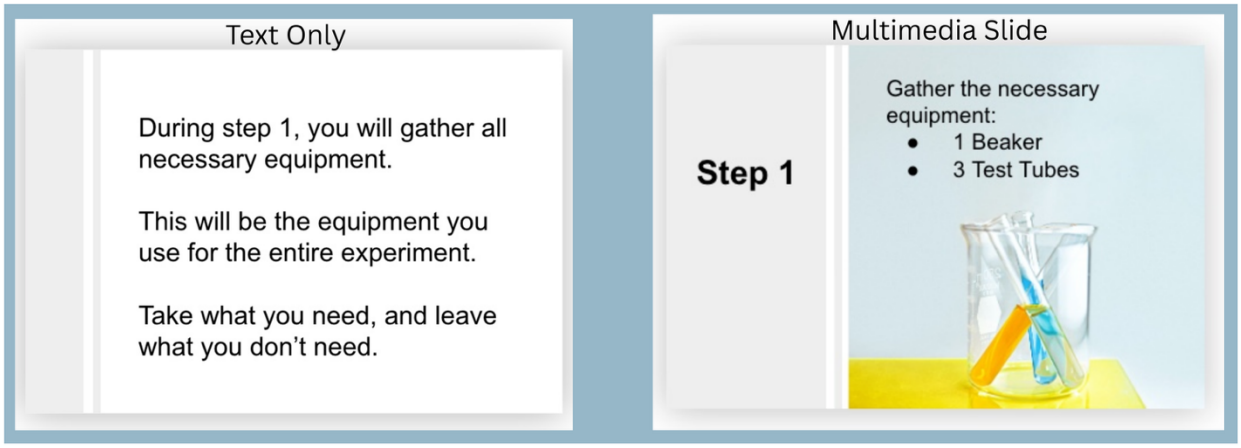
Managing Cognitive Load

Multimedia Principle

The Multimedia Principle is the use of relevant graphics with words to support deeper understanding than words alone (Caskurlu et al., 2021; Clark & Mayer, 2023; Kari Jabbour, 2012). These graphics can include pictures, drawings, photographs, animation, video, or other visual representations, while the words may be written or spoken (Clark & Mayer, 2023; Kari Jabbour, 2012). Relevant graphics can support deeper processing when they work with the text to communicate the instructional message (Clark & Mayer, 2023; Mayer, 2009). In contrast, purely decorative visuals should be minimized to avoid interfering with understanding of the material (Kari Jabbour, 2012). Using words and visuals together can help learners organize new information and relate it to their existing knowledge (Clark & Mayer, 2023; Mayer, 2009). Because prior knowledge can affect the usefulness of multimedia, instructors should also consider learners' knowledge levels, instructional goals, and the purpose of the lesson when determining how to use multimedia (Caskurlu et al., 2021; Kari Jabbour, 2012). Figure 6 compares a text-only slide with a redesigned slide that combines words and relevant graphics.

Figure 6

Comparison of Text-Only and Multimedia Slide Design



Note. Adapted from "Multimedia Principle," in *Module 4 book: Designing PowerPoint presentations for all*, by Thompson Rivers University, n.d., OpenCourses (https://opencourses.tru.ca/mod/book/view.php?chapterid=422&id=1131).

Example

An example of the Multimedia Principle in piano instruction is teaching a student how musical form is organized. When introducing ABA form, the instructor could provide a short, spoken explanation alongside a visual diagram that represents each section of the piece with labeled blocks. Unrelated pictures, animations, or other decorative elements are avoided to help the student focus on the relationship between the explanation and the musical form. Once the student understands the basic structure, the instructor could connect it to familiar pieces the student has previously learned and identify similar patterns in new music. As the student listens to the piece, they can follow the diagram from the opening A section, through the B section, and

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back to A. This gives the student a visual guide to the piece's structure while they hear those changes in the music.

Reference(s)

Caskurlu, S., Richardson, J. C., Alamri, H. A., Chartier, K., Farmer, T., Janakiraman, S., Strait, M., & Yang, M. (2021). Cognitive load and online course quality: Insights from instructional designers in a higher education context. *British Journal of Educational Technology*, *52*(2), 584–605. https://doi.org/10.1111/bjet.13043

Clark, R. C., & Mayer, R. E. (2023). *E-learning and the science of instruction: Proven guidelines for consumers and designers of multimedia learning* (5th ed.). John Wiley & Sons.

Kari Jabbour, K. (2012). Multimedia principle in teaching lessons. *Acta Didactica Napocensia, 5*(4), 11–16.

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Signaling

Signaling is an instructional strategy that uses cues to direct learners' attention to important information and make the organization of material easier to recognize (Alpizar et al., 2020; Clark & Mayer, 2023; Horvath, 2014; van Gog, 2014, as cited in Alpizar et al., 2020). Table 16 presents examples of different types of instructional signals.

Table 16

Types of Signals Used in Instruction

Type of Signal	Example
Text	Bold, underlined, enlarged, or otherwise emphasized words that draw attention to important information
Visual	Arrows, colors, or pictures that point to or emphasize important material
Gestural	Pointing or eye movements that direct attention
Vocal	Changes in vocal emphasis or intonation that highlight important information

Note. Examples are based on Clark and Mayer (2023), Hayes and Reinking (1991), Lin and Atkinson (2011), and Mautone and Mayer (2001).

By guiding attention toward relevant information and clarifying relationships among ideas, signaling can reduce unnecessary cognitive processing and free working-memory resources for selecting, organizing, and integrating new information (Alpizar et al., 2020; Clark & Mayer, 2023; Fiorella & Mayer, 2021; Moreno & Abercrombie, 2010, as cited in Alpizar et al., 2020). However, signaling should be used purposefully, as additional cues do not always improve comprehension (Kriz & Hegarty, 2007).

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Example

Signaling can be used when a student is learning a piece that includes large jumps in the left hand. The instructor could place a small arrow or other visual cue in the music before each jump to draw attention to the upcoming change in hand position. During the lesson, the instructor might also point to the next location as the student approaches it. These cues help the student notice where extra preparation is needed without adding unnecessary information to the score. As the student becomes more comfortable anticipating the jumps, the visual cues can be removed.

Reference(s)

Alpizar, D., Adesope, O. O., & Wong, R. M. (2020). A meta-analysis of signaling principle in multimedia learning environments. *Educational Technology Research and Development, 68*(5), 2095–2119. https://doi.org/10.1007/s11423-020-09748-7

Clark, R. C., & Mayer, R. E. (2023). *E-learning and the science of instruction: Proven guidelines for consumers and designers of multimedia learning* (5th ed.). John Wiley & Sons.

Fiorella, L., & Mayer, R. E. (2021). Principles for reducing extraneous processing in multimedia learning: Coherence, signaling, redundancy, spatial contiguity, and temporal contiguity principles. In R. E. Mayer & L. Fiorella (Eds.), *The Cambridge handbook of multimedia learning* (3rd ed., pp. 185–198). Cambridge University Press. https://doi.org/10.1017/9781108894333.019

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Chunking

Chunking is an instructional strategy that breaks down larger or more complex content into smaller, easier-to-work-with parts (Gorbunova et al., 2025; Martínez-Huamán et al., 2023; van Merriënboer et al., 2003). Instead of presenting all the information at once, content can be introduced gradually, often beginning with simpler material and progressing to more complex material as learners become familiar with it (Sentz et al., 2019). This helps reduce demands on

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working memory, which can handle only a limited amount of information at one time (Clark & Mayer, 2023). Breaking the material into parts gives learners time to work through and organize one section before more information is added (Clark & Mayer, 2023; Sentz et al., 2019). Table 17 summarizes the components of chunking and how they support learning.

Table 17

How Chunking Supports Learning

Chunking Component	What It Involves	Purpose
Divide content	Break larger or complex content into smaller, meaningful, manageable units.	Reduce the amount of information learners encounter at one time.
Sequence chunks	Introduce information gradually, often progressing from simpler to more complex material.	Allow learners to become familiar with one portion before adding complexity.
Process each chunk	Give learners an opportunity to attend to, organize, and integrate the information in the current chunk.	Support meaningful cognitive processing within the limits of working memory.
Build toward the whole	Introduce additional chunks as learners become familiar with earlier material.	Gradually develop understanding of the larger or more complex content.

Note. Based on Clark and Mayer (2023), Gorbunova et al. (2025), Martínez-Huamán et al. (2023), Sentz et al. (2019), and van Merriënboer et al. (2003).

Example

Chunking can be used when introducing students to new musical vocabulary. Rather than presenting a long list of unrelated terms at once, the instructor can organize them into smaller, more meaningful groups based on their musical function. For example, the student could first learn a group of tempo terms such as largo, andante, and allegro. After becoming familiar with those terms, the student could move on to a group of dynamic terms such as piano, mezzo forte, and forte, followed later by articulation terms such as legato and *staccato*. Organizing related terms into smaller groups limits the amount of new information the student must process at one time while helping them recognize the relationships among the terms within each chunk.

Reference(s)

Clark, R. C., & Mayer, R. E. (2023). *E-learning and the science of instruction: Proven guidelines for consumers and designers of multimedia learning* (5th ed.). John Wiley & Sons.

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Visual Cognitive Load/C-R-A-P

Visual Cognitive Load is the mental effort involved in making sense of visual information and organizing it mentally. Learners cannot process an unlimited amount of visual information at once because both working memory and the visual processing channel have limited capacity (Clark & Mayer, 2023; Sweller, 2020; Sweller et al., 2019). When instructional materials contain too much information or are visually complicated, learners may use cognitive resources trying to navigate the presentation rather than understand the content. Clear and consistent layouts, on the other hand, can make important information easier to find and understand (Lohr, 2008, as cited in Yeh & Cheng, 2010; Sweller, 2020; Sweller et al., 2019; Yeh & Cheng, 2010). Table 18 outlines the four C-R-A-P principles of visual design.

Table 18

The C-R-A-P Principles of Visual Design

C-R-A-P Principle	Description
Contrast	Uses noticeable differences between visual elements to distinguish important information and create emphasis.
Repetition	Reuses visual elements, such as formatting, fonts, headings, or other design features, to create consistency and help learners recognize relationships.
Alignment	Places visual elements in intentional relationships with one another so information appears organized and visually connected.
Proximity	Places related information close together while separating unrelated information, helping learners recognize which elements belong together.

Note. C-R-A-P = contrast, repetition, alignment, and proximity. Summarized from Lohr (2008, as cited in Yeh & Cheng, 2010), Williams (2008, as cited in Yeh & Cheng, 2010), and Yeh and Cheng (2010).

Applying these principles can help learners perceive, organize, and interpret visual information more effectively (Lohr, 2008, as cited in Yeh & Cheng, 2010; Yeh & Cheng, 2010). This supports the broader goal of helping learners manage their cognitive processing capacity (Clark & Mayer, 2023).

Example

An example of managing Visual Cognitive Load with C-R-A-P in piano instruction is designing a visual guide for reading notes on the grand staff. If the guide contains too many competing colors, inconsistent fonts, crowded examples, or labels placed far from the notes they describe,

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the student must expend additional mental effort to figure out where to look and how the information is organized. This can increase unnecessary visual cognitive load and make it harder to focus on reading the notes themselves. **Contrast** can make target notes or key information stand out from the rest of the page. **Repetition** can keep fonts, symbols, headings, and formatting consistent, so students do not have to interpret new visual patterns repeatedly. **Alignment** can organize note names, staff examples, and explanations so related information is visually connected. **Proximity** can place each note name close to its matching staff example while separating unrelated information. By reducing unnecessary visual searching and organization, these principles can help the student use more of their limited cognitive resources to process and understand the musical content.

Reference(s)

Clark, R. C., & Mayer, R. E. (2023). *E-learning and the science of instruction: Proven guidelines for consumers and designers of multimedia learning* (5th ed.). John Wiley & Sons.

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Strategies for Student Learning

Scaffolding

Scaffolding is a student learning strategy in which a teacher, peer, or other knowledgeable person provides temporary support so that learners can complete tasks that would otherwise be beyond their current abilities (van de Pol et al., 2010; Wood et al., 1976, as cited in Kim et al., 2019). The support should match the learner's current level of understanding and change as the learner develops greater knowledge and skill (Belland et al., 2013, as cited in Kim et al., 2019; van de Pol et al., 2010). Table 19 provides examples of scaffolding supports and their instructional purposes.

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Table 19

Examples of Scaffolding Supports

Type of support	How it supports the learner
Modeling or demonstration	Shows the learner how to approach or complete part of a task.
Explanation	Clarifies information or a difficult step when additional understanding is needed.
Hints or guiding questions	Gives the learner enough direction to continue without providing the entire solution.
Feedback	Helps the learner recognize what is working and what may need to change.
Worked examples	Provides a model of how a problem or task can be completed.
Pretraining	Introduces important information before the learner begins a more complex task.
Breaking material into smaller parts	Reduces the amount of information or complexity the learner must manage at one time.

Note. Examples of scaffolding supports synthesized from Clark and Mayer (2023) and van de Pol et al. (2010).

As learners become more capable, the instructor removes supports no longer needed and gives learners more responsibility for completing tasks independently (Kim et al., 2019; van de Pol et al., 2010). Cognitive load research reflects a similar progression. Worked examples can support learners when a task is unfamiliar, while that support can be reduced as expertise develops and learners are ready for more complex tasks (Caskurlu et al., 2021; Kalyuga, 2014, as cited in Caskurlu et al., 2021; Kalyuga et al., 2003, as cited in Caskurlu et al., 2021).

Example

An example of scaffolding in piano instruction is helping a beginner learn to read notes on the staff by labeling and color-coding them. At first, the instructor might write in note names and use colors to connect specific staff notes to the correct piano keys. These supports help the student complete note-reading tasks that would otherwise be too difficult independently. As the student becomes more well-versed in note reading, the instructor removes some note labels and colors while keeping cues on notes that remain unfamiliar. If the student begins making errors, some of those supports can be added back and then reduced again as needed. Eventually, the student identifies and plays the notes without the labels or colors.

Reference(s)

Caskurlu, S., Richardson, J. C., Alamri, H. A., Chartier, K., Farmer, T., Janakiraman, S., Strait, M., & Yang, M. (2021). Cognitive load and online course quality: Insights from instructional designers in a higher education context. *British Journal of Educational Technology, 52*(2), 584–605. https://doi.org/10.1111/bjet.13043

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Kim, N. J., Belland, B. R., & Axelrod, D. (2019). Scaffolding for optimal challenge in K–12 problem-based learning. *Interdisciplinary Journal of Problem-Based Learning, 13*(1), Article 3. https://doi.org/10.7771/1541-5015.1712

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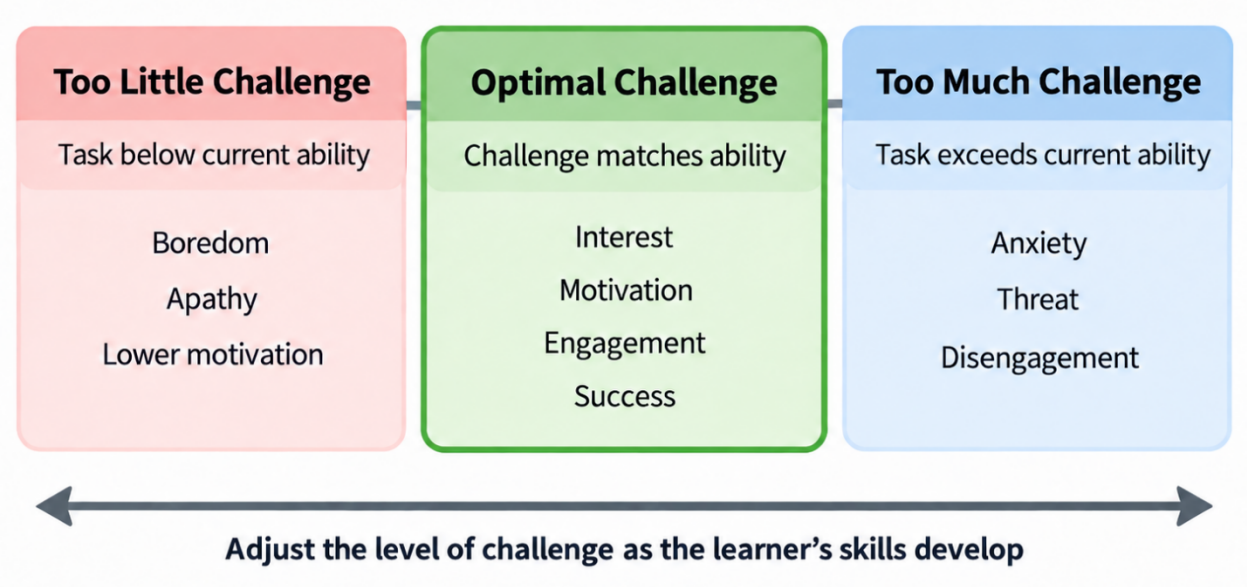
Levels of Challenge

Levels of Challenge is a strategy that sets the difficulty of a learning task at a level appropriate to what the learner already knows and can do (Clarke et al., 2005; Kalyuga, 2009; Soltani et al., 2011). When challenge is appropriately matched, students are more likely to remain interested, motivated, engaged, and successful. Tasks that are too difficult can lead to anxiety or disengagement, while tasks that are too easy can lead to boredom or apathy (Renninger & Hidi, 2015, as cited in Kim et al., 2019; Rheinberg & Vollmeyer, 2003, as cited in Kim et al., 2019; Shernoff, 2013, as cited in Kim et al., 2019; Soltani et al., 2011; Tozman et al., 2015; Willingham, 2009, as cited in Kim et al., 2019). The level of difficulty that works for a learner can change as their knowledge and expertise grow. Tasks should therefore be adjusted to reflect what the learner is ready to do at that point (Caskurlu et al., 2021; Clarke et al., 2005; Kalyuga, 2009). Instruction also needs to account for the limits of working memory. If the material becomes too complex to process at once, its complexity should be reduced so that learners can focus on the essential information (Clark & Mayer, 2023; Fiorella & Mayer, 2022a, as cited in Clark & Mayer, 2023; Mayer, 2021a, as cited in Clark & Mayer, 2023; Mayer, 2022a, as cited in Clark & Mayer, 2023; Paas & Sweller, 2022, as cited in Clark & Mayer, 2023). Figure 7 illustrates the relationship between task difficulty, learner ability, and levels of challenge.

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Figure 7

Levels of Challenge Continuum



Note. Figure created via the author's detailed instructions with assistance from ChatGPT (OpenAI, 2026) to illustrate the relationship between learner skill and task difficulty described by Kim et al. (2019), Soltani et al. (2011), and Tozman et al. (2015).

Example

An example of Levels of Challenge in piano instruction is selecting sight-reading music that is appropriately difficult for a student's current ability. If a student can comfortably sight-read a short piece in a five-finger position with simple rhythms, the instructor might choose a new excerpt that keeps the same hand position but introduces one additional challenge, such as eighth notes. If the student completes the excerpt easily, the instructor can increase the difficulty by selecting music with a hand-position change or a less familiar key. If the student becomes overwhelmed and accuracy begins to break down, the instructor can choose a slightly easier excerpt. In this way, the task's difficulty is adjusted to keep the student sufficiently challenged to develop new skills without making the activity so difficult that it leads to frustration or disengagement.

Reference(s)

Caskurlu, S., Richardson, J. C., Alamri, H. A., Chartier, K., Farmer, T., Janakiraman, S., Strait, M., & Yang, M. (2021). Cognitive load and online course quality: Insights from instructional designers in a higher education context. *British Journal of Educational Technology, 52*(2), 584–605. https://doi.org/10.1111/bjet.13043

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Active Learning

Active learning is a student-centered strategy that engages learners directly in the learning process rather than passively receiving information. It can involve activities such as discussion, problem-solving, collaborative work, peer instruction, and other tasks that require students to actively engage with their learning (Freeman et al., 2014). In active learning, students make sense of new material by relating it to what they already understand instead of simply taking in information (Clark & Mayer, 2023). Active learning can lead students beyond basic recall by giving them opportunities to reason through ideas, apply what they know, and develop stronger conceptual understanding (Clark & Mayer, 2023; Cordray et al., 2009; Freeman et al., 2014; Haukoos & Penick, 1983, as cited in Freeman et al., 2014; Jensen & Lawson, 2011; Martin et al., 2007, as cited in Freeman et al., 2014). Table 20 provides examples of how students actively participate in the learning process.

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Table 20

What Students Do During Active Learning and Why It Matters

Student action	What this can look like
Participate	Students discuss ideas, solve problems, work with peers, use response systems, or complete in-class activities rather than only listening.
Process information	Students identify relevant information, organize it, and connect it with what they already know.
Test and revise ideas	Students use what they already know to approach a new challenge, try possible solutions, and adjust their understanding based on what they discover.
Transfer learning	Students apply knowledge and reasoning to unfamiliar problems rather than repeating information in the same form in which it was taught.
Use higher-level thinking	Students analyze, explain, reason, and apply concepts rather than relying mainly on recall or basic comprehension.

Note. Active-learning examples are drawn from Freeman et al. (2014). Cognitive processing is based on Clark and Mayer (2023). Application, transfer, and adaptive expertise are supported by Cordray et al. (2009), and the distinction between higher- and lower-level learning outcomes is based on Jensen and Lawson (2011).

Example

An example of active learning in piano instruction is having a student examine a recording of their own performance. After recording a short section of a piece, the student listens back, following the score, and identifies specific strengths and areas for improvement, such as rhythm, articulation, dynamics, balance, or phrasing. Rather than immediately receiving corrections from the instructor, the student decides what they would like to change and explains why. They then return to the piano, experiment with those changes, and make a second recording to compare with the first. This keeps the student actively involved in listening, analyzing, making decisions, applying changes, and reflecting on the results rather than simply receiving feedback from the instructor.

Reference(s)

Clark, R. C., & Mayer, R. E. (2023). *E-learning and the science of instruction: Proven guidelines for consumers and designers of multimedia learning* (5th ed.). John Wiley & Sons.

Cordray, D. S., Harris, T. R., & Klein, S. (2009). A research synthesis of the effectiveness, replicability, and generality of the VaNTH challenge-based instructional modules in bioengineering. *Journal of Engineering Education, 98*(4), 335–348. https://doi.org/10.1002/j.2168-9830.2009.tb01031.x

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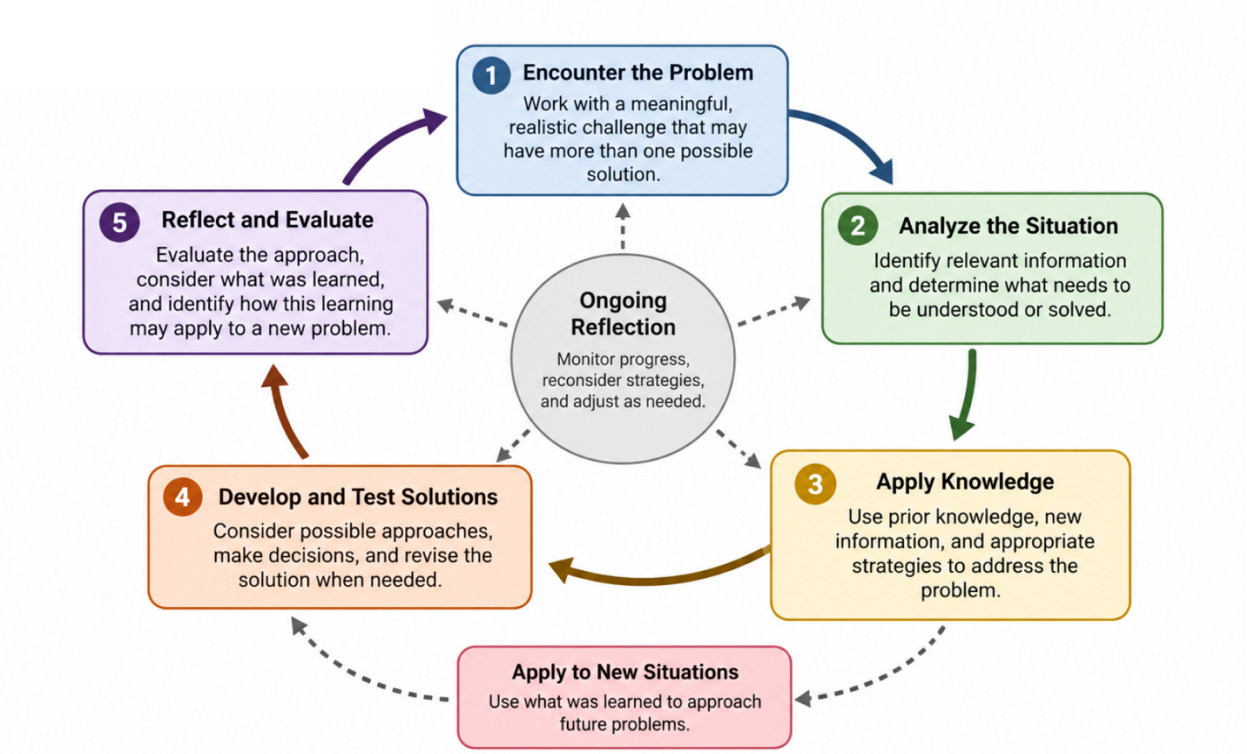
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Problem-Solving Tasks

Problem-Solving Tasks place students in situations that require them to use what they know to work toward a solution. The process involves making sense of the problem, deciding how to approach it, and revising that approach as new information or difficulties emerge (Hmelo-Silver, 2004). Problem-solving activities are especially valuable when they are connected to authentic situations because this encourages students to use knowledge in contexts similar to those in which it may be applied to real-world scenarios (Clark & Mayer, 2023; Cognition and Technology Group at Vanderbilt, 1997, as cited in Hmelo-Silver, 2004; Hmelo-Silver, 2004; Kolodner et al., 2003). Effective problem-solving tasks may be open-ended or allow multiple solutions, and they can be sequenced from simpler to increasingly complex tasks as learners develop greater knowledge and skill (Caskurlu et al., 2021; van Merriënboer et al., 2003; van Merriënboer & Sweller, 2010). The problem-solving tasks can also be varied so students practice applying their knowledge across different situations, which supports flexible understanding and knowledge transfer (Cognition and Technology Group at Vanderbilt, 1997, as cited in Hmelo-Silver, 2004; Hmelo-Silver, 2004; Kolodner et al., 2003; van Merriënboer & Sweller, 2010). A strong problem-solving task gives students a realistic challenge that is difficult enough to require thought, but still within reach. As they work through it, students can reconsider their choices, notice where their approach is working or breaking down, and use that experience when they face a new problem later (Hmelo-Silver, 2004; Kolodner et al., 2003). Figure 8 illustrates a cyclical problem-solving process that incorporates reflection and the application of learning to new situations.

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Figure 8
Problem-Solving Task Cycle



Note. Original figure based on the problem-solving process described by Hmelo-Silver (2004), created by the author with assistance from ChatGPT (OpenAI, 2026).

Example

An example of a Problem-Solving Task in music instruction is having a student compose a counterpoint line to accompany a given melody. The student must determine which notes will create acceptable harmonic intervals with the original melody, while also considering the melodic shape and movement of the new line, rather than relying on a predetermined correct answer. The student can create different solutions if each one follows the established counterpoint guidelines. If a problem arises, such as parallel fifths or a large melodic leap that is not properly resolved, the student identifies the issue and revises the line, considering alternative solutions. As the student becomes more proficient, additional requirements or more complex forms of counterpoint can be introduced. After completing the exercise, the student can evaluate which choices were effective and explain the reasoning behind the revisions.

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Knowing the Learner

Module 4

Learner and Context Analysis

Dick and Carey Model

The Dick and Carey model, also known as the Systems Approach Model, represents a comprehensive and systematic framework for instructional design first introduced in 1978 in "The Systematic Design of Instruction" (Instructional Designers of Penn State, 2018). This influential model stands apart from earlier linear approaches by emphasizing the interconnections between all design elements and incorporating crucial feedback loops throughout the process.

This model's iterative nature makes it particularly effective—instructional designers can continuously refine and modify components based on evaluation data, ensuring the entire instructional system evolves to better meet learner needs. Rather than viewing instruction as simply delivering content, the Dick and Carey model conceptualizes instruction as a complete system aimed at helping learners achieve specific outcomes (Dick et al., 2015). Table 21 outlines the nine steps of the Dick and Carey instructional design model, illustrating how its interconnected components support the systematic development and refinement of instruction.

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Table 21

Nine Steps of the Dick and Carey Instructional Design Model

Steps	Definition/Description
1. Identify Instructional Goals	Determine what learners should be able to do after completing instruction by analyzing needs, examining existing goals, or conducting needs assessments.
2. Conduct Instructional Analysis	Break down the instructional goal into specific component skills and knowledge required for successful performance.
3. Analyze Learners and Contexts	Identify characteristics of target learners, including prior knowledge, skills, attitudes, and the learning environment where skills will be used.
4. Write Performance Objectives	Specify exactly what learners can do, under what conditions, and to what standard after instruction.
5. Develop Assessment Instruments	Create assessments directly tied to performance objectives that measure learner achievement of each objective.
6. Develop Instructional Strategy	Plan the specific instructional activities, including pre-instructional activities, content presentation, learner participation, assessment, and follow-through.
7. Develop and Select Instructional Materials	Create or select instructional materials based on the instructional strategy, including instructor guides, student materials, and media.
8. Design and Conduct Formative Evaluation	Test instructional materials with representative learners to identify areas for improvement before full implementation.
9. Revise Instruction	Use data from formative evaluation to improve the effectiveness of instruction through targeted revisions.

Note. Steps and descriptions are taken from the course-provided Learning Design Toolkit template. The sequence emphasizes the connections among instructional goals, learner analysis, assessment, development, evaluation, and revision.

This comprehensive understanding of learners, including their academic motivation, learning preferences, and contextual needs, enables instructional designers to create targeted, effective learning experiences that align with performance objectives while accommodating the specific characteristics of the learner population.

Example

As an instructional designer at a mid-sized university, I was tasked with developing a new online graduate course in educational leadership. Rather than diving straight into content creation, I employed the Dick and Carey model to ensure a systematic approach.

I met with the subject matter expert to identify the core instructional goal: preparing students to develop data-driven school improvement plans. Through careful analysis, we determined the essential skills students would need and examined the unique characteristics of our working professional student population.

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With established, clear performance objectives, I developed authentic assessments that measure students' ability to analyze school data and create implementation plans. These assessments directly informed my instructional strategy, which balanced theoretical foundations with practical applications relevant to the students' professional contexts.

As I developed the course materials, I maintained focus on the interconnected nature of the Dick and Carey model. When initial testing with a small student group revealed confusion around data analysis procedures, I quickly revised those specific modules while ensuring the changes aligned with the established objectives and assessments.

This systematic, iterative approach allowed me to create a cohesive learning experience that addressed the specific needs of the graduate students while maintaining rigorous academic standards—demonstrating the practical value of the Dick and Carey model in higher education instructional design.

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Sociocultural Factors

Sociocultural Theory

Sociocultural Theory views learning and development as processes molded through participation and interaction with others across diverse social, cultural, and historical environments (Crosslin et al., 2013; Negueruela-Azarola et al., 2025). Rather than separating individual thinking from the social world, SCT proposes that people develop through socioculturally mediated activity, including communication and the use of physical and psychological tools (Negueruela-Azarola et al., 2025). Language plays an important role in SCT because communication with others shapes how individuals understand one another and the world (Negueruela-Azarola et al., 2025). The zone of proximal development (ZPD) focuses on what a learner can do with support compared with what they can do alone. In SCT, mediation can help the learner reach beyond their current independent ability (Negueruela-Azarola et al., 2025).

Lantolf (2024, as cited in Negueruela-Azarola et al., 2025) explains that SCT concepts are interconnected. Negueruela-Azarola et al. (2025) include 46 SCT concepts in their glossary and describe those definitions as starting points rather than complete explanations. Table 22 focuses on several of those concepts.

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Table 22
Key Concepts in Sociocultural Theory

Key Concept	Description
Social, cultural, and historical context	Human learning and development are shaped through participation in social and cultural life and are influenced by the historical contexts in which that activity occurs.
Mediation	Human activity is shaped by the use of culturally developed tools, signs, and other resources. These tools influence how people interact with the world and carry out activities.
Language and communication	Language is a particularly important mediational tool because communication can contribute to interpersonal meaning making and the development of understanding.
Social interaction and joint activity	Development occurs through interaction and participation with other people. Socially mediated communication and interaction contribute to the development of thinking and the construction of meaning.
Internalization	Social processes that first occur between people can be transformed into internal psychological processes, becoming part of an individual's own understanding.
Development	Development involves transformation and growth resulting from participation in sociocultural activities. Although learning and development are related, SCT distinguishes between the two processes.
Zone of Proximal Development (ZPD)	The ZPD describes the difference between what a learner can accomplish independently and what they can accomplish with mediation.
Interaction with more knowledgeable or experienced others	Interaction with someone who has greater knowledge or experience can provide support that helps a learner build new knowledge, gain experience, or accomplish something they could not yet do independently.
Higher psychological functions	Complex forms of thinking, including voluntary attention, self-regulation, intentional memory, planning, categorization, and mediated imagination, develop through sociocultural mediation.
Activity	Learning and development occur through culturally organized activity involving individuals, other people, goals, tools, and broader social structures.

Note. Concepts and descriptions are synthesized from Crosslin et al. (2013) and Negueruela-Azarola et al. (2025). The table presents selected key concepts rather than a comprehensive account of Sociocultural Theory.

Example

An example of Sociocultural Theory in piano instruction is two students working together to learn a piano duet. Although each student has a separate part, they must communicate and cooperate to create a unified performance. They may discuss tempo, rhythm, phrasing, and

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dynamics as they learn to listen to and respond to one another. When a difficult rhythm causes them to fall out of sync, the student with a stronger understanding can explain how to count it and help the other practice. They might also disagree about how a passage should sound. For example, one student might suggest adding a short musical run, while the other feels it interferes with their melody. After experimenting with different options, they find a version that allows the creative idea to be injected while maintaining the melody. Their final interpretation reflects both students' contributions and the evolution of their ideas as they listen to and learn from one another.

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Universal Design for Learning

Universal Design for Learning (UDL) is an evidence-based educational design framework developed by CAST that applies principles of universal design and research on human learning to create flexible, accessible learning experiences for all learners (Bastoni et al., 2023; Fan, 2023; Meyer et al., 2014, as cited in Bastoni et al., 2023; Rose & Meyer, 2002, as cited in Bastoni et al., 2023). UDL is based on the understanding that learners naturally vary in their backgrounds, strengths, needs, identities, motivation, and ways of processing and interacting with information, so a single instructional approach will not work equally well for everyone (Bastoni et al., 2023; CAST, 2024; Rose & Meyer, 2002, as cited in Moore, 2007). Rather than locating learning difficulties only within the learner, UDL recognizes that barriers can arise through interactions with inflexible materials, methods, technologies, and learning environments (Moore, 2007; Rose & Meyer, 2002, as cited in Moore, 2007). To reduce these barriers, UDL emphasizes proactive, intentional design from the outset of instruction, with clear learning goals and flexible methods, materials, supports, technologies, and opportunities for learners to demonstrate their knowledge (Bastoni et al., 2023; Fan, 2023; Moore, 2007; Rose & Meyer, 2002, as cited in Moore, 2007). The framework is organized around multiple means of engagement, representation, and action and expression, which support differences in how learners engage with learning, access and understand information, and act on or communicate their learning (Bastoni et al., 2023; CAST, 2024). CAST (2024) identifies the goal of UDL as developing learner agency that is “purposeful & reflective,” “resourceful & authentic,” and “strategic & action-oriented” (“The UDL Guidelines” section, para. 2). Table 23 shows the components of Universal Design for Learning.

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Table 23

Components of Universal Design for Learning (UDL)

UDL Principle	CAST 3.0 Guidelines	Purpose and Alignment With UDL
Multiple Means of Engagement	**Access:** Welcoming Interests & Identities	**Access:** Reduces barriers to engagement by recognizing learners' interests, identities, and experiences, providing meaningful choices, and encouraging participation.
	Support: Sustaining Effort & Persistence	**Support:** Supports continued motivation and engagement through meaningful goals, appropriate challenge, collaboration, belonging, and constructive feedback.
	Executive Function: Emotional Capacity	**Executive Function:** Develops learners' awareness of emotions, motivations, and beliefs, supporting reflection, empathy, and their ability to navigate emotional aspects of learning.
Multiple Means of Representation	**Access:** Perception	**Access:** Reduces barriers to accessing information by providing flexible ways for learners to perceive instructional content, including accessible formats and adjustable presentations.
	Support: Language & Symbols	**Support:** Supports understanding by clarifying vocabulary, symbols, and language structures and providing multiple ways to interpret information.
	Executive Function: Building Knowledge	**Executive Function:** Develops understanding by connecting new information to prior knowledge, identifying patterns and relationships, constructing meaning, and transferring learning to new situations.
Multiple Means of Action & Expression	**Access:** Interaction	**Access:** Reduces barriers to participation by providing accessible ways to interact with materials, navigate learning environments, and respond using different methods and technologies.
	Support: Expression & Communication	**Support:** Supports learners in demonstrating their knowledge through multiple forms of communication, expression, and creative work, with appropriate opportunities for supported practice.
	Executive Function: Strategy Development	**Executive Function:** Develops learners' ability to establish goals, plan for challenges, organize resources, use effective strategies, and monitor their own progress toward learning goals.

Note. The three UDL principles and their associated guidelines are drawn from CAST (2024). Access focuses on reducing barriers to learning opportunities, Support facilitates the learning process, and Executive Function develops learners' capacity for intentional and self-directed learning. The descriptions are paraphrased from CAST (2024). Together, these components support the development of learner agency.

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Example

An example of Universal Design for Learning (UDL) in piano instruction is teaching students how sharps, flats, and naturals affect musical pitches. The instructor establishes a common learning goal: students will recognize these symbols and understand how they change the notes being played. Before the lesson, the instructor prepares enlarged sheet music, a keyboard diagram, and a few musical demonstrations so students can compare the written symbols with the sounds they produce. Students can also look for examples of these symbols in pieces they already enjoy playing. To show their understanding, some students prefer to play the altered notes on their piano, while others might identify the corresponding keys on a diagram or explain how the symbols change the pitches. Regardless of how they respond, students are working toward the same learning goal. The instructor also asks which resources were most helpful so they can give students a voice in what they prefer for future lessons. Since these options are considered in lesson planning, students have multiple ways to access the material and demonstrate their understanding without waiting until they encounter difficulties.

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